Offline and Preference-Based Reinforcement Learning for Robotics: A Comprehensive Survey

Abstract

Reinforcement Learning (RL) has fundamentally transformed the landscape of autonomous sequential decision-making, equipping robotic agents with the capacity to acquire complex motor skills and cognitive behaviors. However, deploying RL in physical robotics presents two formidable bottlenecks: the prohibitive cost and safety risks of online environmental interaction, and the extreme difficulty of engineering dense, well-shaped, and human-aligned reward functions. To resolve the former, Offline Reinforcement Learning has emerged to train policies entirely from static, pre-collected datasets. To address the latter, Preference-Based Reinforcement Learning (PbRL) eliminates the need for manual reward design by learning reward models from pairwise human judgments. The intersection of these two paradigms—Offline Preference-Based Reinforcement Learning (Offline PbRL)—represents a frontier in modern artificial intelligence, promising to yield highly capable, human-aligned robotic agents without requiring active, real-world trial and error.

This comprehensive survey systematically reviews the theoretical foundations, algorithmic developments, and practical applications of Offline PbRL in robotics. We detail the evolution from two-phase reward-modeling architectures to one-phase, reward-free paradigms such as Direct Preference Optimization (DPO) and Inverse Preference Learning (IPL). We further analyze advanced credit assignment mechanisms, the integration of Vision-Language-Action (VLA) foundation models, and the deployment of multi-modal generative feedback. Finally, we examine the persistent challenges of data corruption, adversarial robustness, and multi-agent scalability, proposing a roadmap for future research in building scalable, safe, and robust robotic systems.

Introduction

Achieving robust and generalizable performance in real-world robotics remains a fundamental challenge, particularly in manipulation and locomotion tasks that require high-dimensional, continuous control [1], [2]. While online deep reinforcement learning (DRL) has demonstrated remarkable success in simulated environments, transferring these successes to physical robots is encumbered by the "reality gap" and the sheer sample complexity of trial-and-error learning [3], [4]. In physical environments, arbitrary exploration can result in catastrophic hardware damage or unsafe interactions with humans [5], [6]. Furthermore, designing a numerical reward function that accurately encapsulates complex human intent without falling victim to "reward hacking"—where an agent exploits loopholes in the reward specification to achieve high scores without completing the intended task—is notoriously difficult [7], [8], [9].

To circumvent the interactive data collection bottleneck, Offline Reinforcement Learning

(Offline RL) has gained prominence [1], [10]. Offline RL allows agents to learn policies entirely from fixed, pre-collected datasets (e.g., prior sub-optimal rollouts, human demonstrations, or random exploration) without requiring further environmental interaction [11], [12]. While this eliminates the risks associated with online exploration, it introduces severe algorithmic challenges, primarily distributional shift and overestimation bias [1], [13]. Because the agent cannot explore to correct its value estimates, querying out-of-distribution (OOD) actions leads to compounded errors during temporal difference (TD) bootstrapping [13], [14].
Concurrently, Preference-Based Reinforcement Learning (PbRL)—also referred to as Reinforcement Learning from Human Feedback (RLHF)—has emerged as a robust solution to the reward engineering problem [15], [16], [17]. Instead of relying on a hard-coded scalar reward, PbRL algorithms query human evaluators to express a preference between pairs of trajectory segments [8], [18]. These pairwise preferences are subsequently used to train a surrogate reward model that aligns with human intuition [19], [20]. While highly successful in aligning Large Language Models (LLMs) [17], [21], traditional PbRL relies heavily on online environment interactions to gather new trajectory pairs for evaluation, making it sample-inefficient and unsuitable for direct robotic deployment [22], [23].
The synthesis of these domains, Offline PbRL, offers a compelling framework: learning human-aligned control policies strictly from pre-collected offline datasets annotated with preference labels [24], [25], [26]. However, this synthesis inherits the vulnerabilities of both parent domains. Learning step-wise reward functions from sparse trajectory-level preferences induces severe credit assignment issues [27], [28]. Furthermore, reward estimation errors are exacerbated by the pessimism mechanisms inherent to offline RL [29], [30].
This survey unpacks the algorithmic landscape of Offline PbRL for robotics. We analyze foundational frameworks, the transition from explicit reward modeling to direct policy optimization (e.g., DPO, IPL), advanced techniques for spatial and temporal credit assignment, and the integration of large-scale Vision-Language-Action (VLA) architectures. We comprehensively review the theoretical guarantees, algorithmic constraints, and prevailing challenges shaping the future of autonomous, human-aligned robotics.

Methods

The methodological landscape of Offline PbRL can be broadly categorized into several algorithmic families, characterized by how they estimate rewards, optimize policies, and manage out-of-distribution data.

Foundations: Offline RL and the Preference MDP

The standard reinforcement learning framework is formulated as a Markov Decision Process (MDP) defined by the tuple $\mathcal{M} = \langle \mathcal{S}, \mathcal{A}, \mathcal{T}, r, \rho_0, \gamma \rangle$, where $\mathcal{S}$ is the state space, $\mathcal{A}$ is the continuous or discrete action space, $\mathcal{T} : \mathcal{S} \times \mathcal{A} \times \mathcal{S} \rightarrow [0, 1]$ represents the transition dynamics, $r : \mathcal{S} \times \mathcal{A} \rightarrow \mathbb{R}$ is the reward function, ρ_0 is the initial state distribution, and $\gamma \in (0, 1)$ is the discount factor [31], [32]. The objective is to learn a policy

$\pi(a|s)$ that maximizes the expected cumulative discounted return $J(\pi) = \mathbb{E}_{\tau\sim\pi}[\sum_{t=0}^{\infty} \gamma^t r(s_t, a_t)]$.

In Offline RL, the agent is deprived of the ability to interact with the environment. Instead, it is provided with a static dataset $\mathcal{D}_{\text{off}} = \{(s_i, a_i, r_i, s'_i)\}_{i=1}^N$ collected by unknown behavior policies π_β [11], [31]. Standard off-policy RL algorithms fail catastrophically in this setting due to overestimation bias. When the policy evaluation step queries the Q-function for actions not present in $\mathcal{D}_{\text{off}}$, the function approximator yields erroneously high values, which are then compounded across the horizon via Bellman updates [10], [13]. To mitigate this, offline RL algorithms enforce conservatism. Approaches like Conservative Q-Learning (CQL) actively penalize the Q-values of unseen actions, while Implicit Q-Learning (IQL) restricts the policy improvement step to actions strictly within the support of the dataset, effectively performing a weighted form of Behavior Cloning (BC) [1], [13].

In domains where designing $r(s, a)$ is intractable, PbRL redefines the MDP into a Preference MDP (MDPP) [8], [33]. The agent is not given absolute numerical rewards but instead receives comparative judgments over trajectories. Given a pair of trajectory segments σ_0 and σ_1, an expert provides a preference label $y \in \{0, 1, 0.5\}$, where $y = 1$ denotes $\sigma_1 \succ \sigma_0$ (σ_1 is preferred) [18], [24]. To translate these discrete labels into a differentiable objective, PbRL commonly utilizes the Bradley-Terry (BT) model, framing the probability of a preference as a softmax function over the latent cumulative rewards of the segments [8], [19]:

$$P_\psi[\sigma_1 \succ \sigma_0] = \frac{\exp(\sum_t \hat{r}_\psi(s_t^1, a_t^1))}{\exp(\sum_t \hat{r}_\psi(s_t^0, a_t^0)) + \exp(\sum_t \hat{r}_\psi(s_t^1, a_t^1))}$$

Here, $\hat{r}_\psi$ is a neural network parameterized by ψ that predicts step-wise rewards. The reward model is trained by minimizing the binary cross-entropy loss between the predicted probabilities and the ground-truth human labels across a preference dataset $\mathcal{D}_{\text{pref}}$ [18], [32].

The Two-Phase Pipeline and Reward Labeling

Offline PbRL traditionally executes a two-phase architecture strictly on pre-recorded data [25], [26], [34]. In the first phase, a preference dataset $\mathcal{D}_{\text{pref}}$ is used to train a step-wise reward model $\hat{r}_\psi$ using the BT model. In the second phase, the trained reward model annotates a larger, reward-free offline dataset $\mathcal{D}_{\text{off}}$ with scalar reward proxies, and a conservative offline RL algorithm (e.g., CQL, IQL) is trained on this pseudo-labeled dataset to extract the optimal policy [26], [30].

While conceptually elegant, this two-phase methodology suffers from severe compounded errors. The reward model often overestimates the rewards for specific sub-optimal behaviors due to spurious correlations in the limited preference dataset. When this biased reward model relabels the offline dataset, it induces optimistic trajectory stitching—the offline RL algorithm erroneously pieces together sub-optimal transitions, believing them to yield high rewards, thereby undermining the pessimistic mechanisms required for offline stability [29], [30].
To mitigate these issues, frameworks such as Binary Reward Labeling (BRL) bypass continuous reward modeling entirely. BRL maps preference feedback into strict binary discrete labels (+1 for preferred, -1 for dispreferred), simplifying the optimization landscape for the downstream offline agent and preventing magnitude-based overestimation [35]. Alternatively, In-Dataset Trajectory Return Regularization (DTR) integrates Conditional Sequence Modeling (CSM) with Temporal Difference Learning (TDL). By utilizing a Decision Transformer to anchor the policy to valid behavior policy returns, DTR filters out overestimated rewards and mitigates the risk of learning inaccurate trajectory stitching under reward bias [29], [30].

Feature	**Standard Offline RL**	**Online PbRL**	**Offline PbRL (Two-Phase)**
Data Source	Static labeled datasets	Online interactions	Static preference + unlabeled data
Reward Signal	Ground-truth numerical	Learned via human queries	Learned via offline preference data
Primary Challenge	Distributional shift (OOD)	Query efficiency & Safety	Reward bias + OOD compounding
Safety in Robotics	High (No active exploration)	Low (Requires environment trials)	High (Fully offline)

Direct Policy Optimization and Reward-Free Frameworks

The inherent instability of the two-phase pipeline—where errors in the reward model irreparably damage the subsequent RL policy—has catalyzed a paradigm shift toward one-phase, reward-free optimization frameworks [28], [36], [37].
The most prominent of these is Direct Preference Optimization (DPO) [38], [39]. DPO leverages a mathematical mapping between reward functions and optimal policies under a KL-divergence constraint. By recognizing that the optimal policy π^* for a given reward function can be expressed in closed form relative to a reference policy π_{ref}, DPO reformulates the Bradley-Terry preference model directly in terms of the policy likelihoods [38], [40]. The

DPO loss function entirely bypasses explicit reward modeling, optimizing the policy π_θ using a simple binary cross-entropy objective:

$$\mathcal{L}_{\text{DPO}}(\pi_\theta; \pi_{\text{ref}}) = -\mathbb{E}_{(x,y_w,y_l)\sim\mathcal{D}}\left[\log\sigma\left(\beta\log\frac{\pi_\theta(y_w|x)}{\pi_{\text{ref}}(y_w|x)} - \beta\log\frac{\pi_\theta(y_l|x)}{\pi_{\text{ref}}(y_l|x)}\right)\right]$$

where y_w and y_l denote the preferred and dispreferred trajectories, respectively, and β controls the strength of the KL penalty [38], [40].

While highly stable, standard DPO relies on a static temperature parameter β, which can lead to suboptimal training on highly diverse robotic datasets. Consequently, numerous variants have been proposed to adapt the framework for complex control tasks:

- **Margin-Adaptive DPO (MADPO):** Utilizes a localized reward model to estimate preference margins, applying a continuous, adaptive weight to the DPO loss for each training sample to prevent overfitting on easy examples and under-learning from informative ones [41], [42].
- ε **-DPO:** Dynamically adjusts the KL penalty for each preference pair based on the monotonicity of logits, preventing excessive deviation while maximizing confidence [43], [44].
- **Entropy Controllable DPO (H-DPO):** Introduces regularizers to prevent mode-seeking collapse and enhance the distributional sharpness of the resulting policy, which is critical for continuous robotic control [45], [46].
- **Distributionally Robust DPO (WDPO / KLDPO):** Addresses preference distribution shifts by employing Wasserstein and Kullback-Leibler uncertainty sets, ensuring robust performance across varying preference models [47], [48].
- **EM-DPO / MinMax-DPO:** Utilizes Expectation-Maximization to cater to heterogeneous human annotators by clustering users into latent preference types and optimizing a mixture of models [40].

An alternative one-phase approach specifically tailored for offline RL is Inverse Preference Learning (IPL) [49], [50]. IPL leverages the insight that, under a fixed policy, the Q-function encodes all necessary information about the reward function. Rather than learning an explicit reward neural network, IPL optimizes the implicit rewards induced by the learned Q-function to be consistent with expert preferences [49], [51]. This drastically reduces the parameter count and algorithmic hyperparameters, achieving competitive performance on continuous control robotics benchmarks without relying on computationally expensive transformer-based reward models [50], [52].

Further advancing this direction, Preference-based Value Optimization (PVO) and Adversarial Preference-based Policy Optimization (APPO) formulate offline PbRL as a two-player adversarial game between the policy and a model. This formulation eliminates the need for intractable explicit confidence set constructions while enforcing pessimism in a tractable manner, maintaining rigorous theoretical sample complexity bounds [53], [54], [55]. Additionally, PbRL with Constrained Actions (PRC) limits the reinforcement learning agent to optimize over a constrained action space that excludes out-of-distribution state-actions, mitigating the risk of reward hacking at the policy level [56], [57].

Addressing Spatial and Temporal Credit Assignment

A critical vulnerability of applying standard Bradley-Terry preference models to robotic trajectories is the spatial and temporal credit assignment problem [27], [58]. When a human evaluator expresses a preference for a 100-step trajectory over another, it is rarely because all 100 steps were superior; rather, specific decisive moments (e.g., successfully grasping an object) dictate the preference. Standard cross-entropy minimization tends to smear this reward evenly across all states in the segment, leading to flat, uninformative reward landscapes [27].

To address temporal misalignment, Preference Learning with Advantage-Weighted Segments (PAWS) restructures policy optimization to occur at the segment level [59]. Rather than forcing the model to infer step-wise rewards, PAWS learns an advantage function evaluated directly over trajectory segments, employing trust-region constrained updates to maintain stability without relying on unreliable single-step assignments [59].

Conversely, Hindsight Preference Learning (HPL) tackles the assumption of Markovian rewards [60], [61], [62]. HPL posits that human evaluators judge actions based on their future outcomes, not just immediate states. HPL incorporates a conditional Variational Autoencoder (VAE) to learn a compact latent representation of future trajectories. The reward function is then conditioned on this hindsight information, marginalizing over potential future outcomes to generate highly robust reward signals that align better with holistic human cognition [61], [63].

To bridge the gap between fine-grained spatial supervision and coarse preferences, Search-Based Preference Weighting (SPW) unifies imitation learning and PbRL [27], [58], [64]. SPW requires an auxiliary, albeit small, dataset of expert demonstrations. For every transition in a preference-labeled trajectory, SPW executes a k-nearest neighbor search against the expert demonstrations, assigning high importance weights to states that closely resemble expert behavior [27], [58]. This provides step-wise density to the otherwise sparse preference labels, guiding the reward model to focus on decisive transitions.

Advanced Reward Architectures

Beyond structural credit assignment, enhancing the representational capacity of the reward model itself is a crucial area of research. Standard Multi-Layer Perceptrons (MLPs) often fail to capture the multi-modal nature of complex robotic tasks.

Diffusion Preference-based Reward (DPR) models discard standard MLPs in favor of diffusion models to map state-action pairs to complex preference distributions [31], [65]. By modeling preference distributions directly through the iterative denoising process of a diffusion model, DPR yields highly expressive reward landscapes capable of handling multi-modal behaviors, stabilizing the downstream offline RL training [31], [66].

Another hybrid approach is the use of Residual Reward Models (RRM) [67], [68]. RRM leverages prior knowledge—often in the form of an imperfect heuristic reward or an Inverse Reinforcement Learning (IRL) baseline extracted from demonstrations—and trains a residual reward network specifically on human preferences to correct the prior where it deviates from preferred behavior. This decomposition accelerates convergence and improves robustness, as the agent is guided by the prior reward during early training while being fine-tuned by the preference residual [67], [68].

Credit Assignment Algorithm	**Primary Mechanism**	**Addressed Limitation**
PAWS	Advantage-weighted segment updates	Temporal smearing of step-wise rewards
HPL	VAE-encoded future trajectory conditioning	Markovian reward assumption
SPW	k-NN search against expert demonstrations	Spatial sparsity in preference labels
DPR	Diffusion-based distribution modeling	MLP capacity limits for multi-modal tasks
RRM	Heuristic prior + preference residual	Slow convergence of ab initio reward learning

VLA Post-Training and Multimodal Feedback

As robotic systems scale toward general-purpose applications, the integration of Vision-Language-Action (VLA) foundation models has become a dominant trend [69], [70]. VLAs map high-dimensional visual observations and linguistic instructions directly to continuous or discretized robot actions [69], [71]. However, post-training alignment for VLAs faces unique architectural hurdles not present in LLMs.

VLAs exist in two primary architectural paradigms: autoregressive (discrete-token prediction, e.g., OpenVLA) and flow-matching (continuous action chunking, e.g., π_0) [69], [72]. While applying DPO to autoregressive models is straightforward by mapping action bins to token probabilities, flow-matching models lack closed-form per-chunk log-probabilities, which previously precluded DPO application [69], [70]. Frameworks like CrossVLA circumvent this by introducing a surrogate flow-matching log-probability estimator based on negative Mean Squared Error (MSE), enabling stable, cross-paradigm preference alignment for both discrete and continuous VLAs [69], [72].

Further extending VLA post-training, Action Preference Optimization (APO) and Human-assisted Action Preference Optimization (HAPO) enable VLAs to learn from sub-optimal human interventions and deployment failures [73], [74], [75]. Because robotic interaction is irreversible (unlike text generation), generating perfect paired negative/positive rollouts is impossible. APO utilizes an adaptive reweighting algorithm that translates binary desirability signals from human corrections into the discrete action token space. This dynamic modulation addresses the token probability mismatch, allowing the VLA to suppress failure-prone actions and generalize to out-of-distribution environments [73], [74]. Similarly, Reinforcement Interactive Post-Training (RIPT-VLA) introduces dynamic rollout sampling to discard ambiguous contexts and isolate high-advantage learning signals, dramatically boosting few-shot learning and interactive task success [76].

Simultaneously, the burden of human labeling is being alleviated by employing Vision-Language Models (VLMs) as automated preference annotators [26], [77], [78]. Frameworks such as PRIMT and RL-VLM-F query pre-trained foundation models (e.g., GPT-4V) with paired visual frames of a robot's execution and a textual goal description, instructing the VLM to return a preference label [24], [77], [79]. PRIMT specifically utilizes a hierarchical neuro-symbolic fusion strategy, synthesizing the complementary strengths of both VLMs and LLMs to overcome query ambiguity and provide reliable synthetic feedback for complex locomotion and manipulation tasks [78], [80]. Furthermore, Preference-based Action Representation Learning (PbARL) utilizes mutual information action encoders to decouple common task structures from user-specific preferences, allowing robots to personalize behaviors without degrading foundational pre-trained capabilities [81], [82].

Challenges and Prospects

Despite the rapid algorithmic maturation of Offline PbRL, transitioning these systems into ubiquitous real-world deployment requires overcoming severe statistical, systemic, and security challenges.

Data Corruption, Bias, and Adversarial Robustness

Because Offline PbRL relies heavily on human evaluators (or VLM proxies), the resulting preference datasets are inherently plagued by noise, cognitive bias, and adversarial corruption [83], [84], [85]. Evaluators experience fatigue, shifting expectations over time, and varied interpretations of tasks, leading to contradictory preference labels [86]. Furthermore, in crowdsourced environments, adversarial annotators may intentionally flip labels to poison the reward model [84], [87].

Research into corruption-robust Offline PbRL has established that standard algorithms fail catastrophically even at low corruption rates (e.g., an ε -fraction of 10% label flipping) [83], [85], [88]. To counteract this, methodologies like the Trust Parameters (TTP) framework jointly learn a shared reward model alongside expert-specific trust parameters. During gradient optimization, these parameters naturally evolve to identify and invert systematically adversarial preferences, recovering useful signals from malicious actors rather than merely discarding them [84].

In the realm of direct alignment, DPO has been shown to be highly vulnerable to targeted label-flip attacks. Analytical frameworks have demonstrated that DPO's log-linear gradient shifts predictably under flipped labels, allowing adversaries to execute Binary-Aware Lattice

Attacks (BAL-A) or Binary Matching Pursuit Attacks (BMP-A) to poison the optimal policy [87], [89]. Conversely, Fairness-Aware Offline RLHF (Fair-RLHF) utilizes causal Directed Acyclic Graphs (DAGs) to identify unfair causal paths in human feedback (e.g., demographic or behavioral biases) and applies Pareto-efficient fairness constraints to ensure that the resulting robotic policies behave ethically and equitably [90], [91]. The continued development of benchmarks like B-Pref, which explicitly simulate varying degrees of human irrationality and adversarial behavior, remains critical for standardizing robustness evaluation [92], [93].

Scalability, Query Efficiency, and Out-of-Distribution Generalization

Even within fully offline settings, the query efficiency of PbRL remains a bottleneck. Training accurate reward models typically requires thousands of preference pairs [51], [52]. If the provided dataset lacks adequate coverage, the resulting policy will fail to generalize [1], [13].
To maximize the informativeness of the limited preference data, algorithms like OPRIDE (Offline PbRL via In-Dataset Exploration) employ principled exploration strategies to select trajectory pairs that yield the highest information gain by analyzing value differences between trajectories [94], [95]. By integrating discount scheduling to mitigate the overoptimization of learned rewards, OPRIDE drastically reduces the required human queries [94]. Similarly, LEASE (offLine prEference-bAsed RL with high Sample Efficiency) utilizes learned transition models to generate synthetic, unlabeled trajectories. It then applies uncertainty-aware selection mechanisms (measuring the variance of probabilities across multiple reward models) to generate high-confidence pseudo-labels, augmenting the training corpus without requiring online interaction [96], [97], [98]. Video-Based Optimal Transport Preference (VOTP) extends this logic by leveraging optimal transport to align visual trajectories within the representation space of Video Foundation Models (ViFMs), generating high-fidelity pseudo-labels for massive unlabeled video datasets [99], [100].

Multi-Agent Preference Learning (MARL)

While single-agent Offline PbRL has been thoroughly explored, extending these paradigms to Multi-Agent Reinforcement Learning (MARL) introduces exponential complexity [101], [102], [103]. In cooperative multi-agent settings (e.g., swarm robotics, multi-arm assembly), the joint state-action space grows exponentially, making traditional reward modeling computationally intractable [101], [104]. Furthermore, assigning credit to a specific agent's action based on a single trajectory-level preference query is highly ambiguous [102].
Frameworks such as O-MAPL (Offline Multi-Agent Preference Learning) circumvent these issues by adopting an end-to-end learning approach that leverages the Centralized Training with Decentralized Execution (CTDE) paradigm [102], [103], [104]. O-MAPL utilizes a multi-agent value decomposition strategy to factorize the soft Q-function directly from preference data, preserving convexity and ensuring local-global consistency without relying on an explicit reward model [102], [103]. Furthermore, integrating LLMs to act as centralized preference adjudicators for multi-agent trajectories is a highly active area of research, leveraging progressive trajectory labeling to partition unlabeled data into expert and non-expert trajectories prior to value decomposition [101], [105].

Sim-to-Real Transfer and Continual Learning

The transfer of offline-trained policies to dynamic real-world environments is heavily contingent on bridging the reality gap. Frameworks are increasingly leveraging Off-Policy Evaluation (OPE) to automatically select optimal state representations and reward functions exclusively from logged data, bridging the sim-to-real gap without necessitating hazardous online policy testing [106]. Furthermore, Continual Reinforcement Learning (CRL) explores how offline PbRL agents can adapt to non-stationary environments and lifelong adaptation scenarios, requiring a triangular balance among plasticity (acquiring new skills), stability (avoiding catastrophic forgetting of human preferences), and scalability [107].

Conclusion

The intersection of Offline and Preference-Based Reinforcement Learning represents a paradigm shift in autonomous robotics, moving the field away from the rigid constraints of handcrafted rewards and the perilous inefficiencies of online exploration. By leveraging static datasets and pairwise human or VLM-generated judgments, Offline PbRL enables the extraction of highly capable, aligned control policies suitable for real-world deployment.

The transition from brittle, two-phase reward modeling pipelines to elegant, one-phase optimization frameworks like Direct Preference Optimization (DPO) and Inverse Preference Learning (IPL) has dramatically improved the mathematical stability and computational efficiency of the field. Coupled with advanced credit assignment mechanisms—such as Hindsight Preference Learning (HPL) and Search-Based Preference Weighting (SPW)—agents can now glean fine-grained control signals from coarse, trajectory-level human feedback. Furthermore, the integration of these techniques into Vision-Language-Action (VLA) foundation models establishes a highly scalable pathway toward general-purpose, instruction-following robotics.

However, as these systems scale, addressing the vulnerabilities inherent in subjective human data is paramount. Future research must prioritize adversarial robustness, temporal consistency in reward evaluation, query efficiency through generative data augmentation, and scalable multi-agent alignment architectures. By continuing to refine the statistical foundations of preference extraction and pessimistic policy optimization, the robotics community can unlock the full potential of human-aligned artificial intelligence in complex, unstructured physical environments.

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